

ASSIGNMENT 1

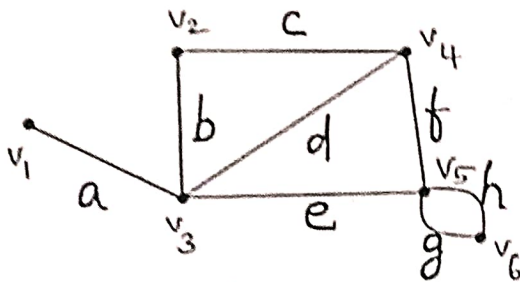
- 1) Consider the graph G with vertex set $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $E = \{(v_1, v_2), (v_1, v_3), (v_2, v_4), (v_2, v_5), (v_3, v_5), (v_3, v_6), (v_4, v_7), (v_5, v_7), (v_6, v_7)\}$.

- (a) Draw the graph G .
- (b) Write the adjacency matrix $X(G)$ with vertex labels.
- (c) Find the degree of each vertex.
- (d) Find the incidence matrix $A(G)$.

- 2) Consider the graph G with vertex set $V = \{v_1, v_2, v_3, v_4, v_5\}$ and edge set $E = \{a = (v_1, v_2), b = (v_2, v_3), c = (v_3, v_5), d = (v_1, v_4), e = (v_4, v_5), f = (v_2, v_4)\}$.

- (a) Draw the graph G .
- (b) List all distinct paths between vertices v_1 and v_5 .
- (c) Construct the path matrix $P(v_1, v_5)$.
- (d) Identify any edge that lies in every path and any edge that lies in no path between v_1 and v_5 .

- 3) Define Circuit Matrix. Find the circuit matrix of the following graph.



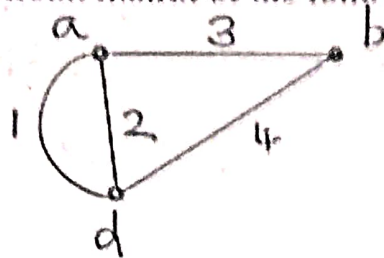
- 4) Consider a graph G with vertex set $V = \{v_1, v_2, v_3, v_4, v_5\}$ and edge set $E = \{a = (v_1, v_2), b = (v_2, v_3), c = (v_3, v_5), d = (v_1, v_4), e = (v_4, v_5), f = (v_2, v_4)\}$

- (a) Draw the graph of G
- (b) List all distinct paths between vertices v_1 and v_5
- (c) Construct the path matrix $P(v_1, v_5)$

- (d) Identify any edge that lies in every path and any edge that lies in no path between v_1 and v_5

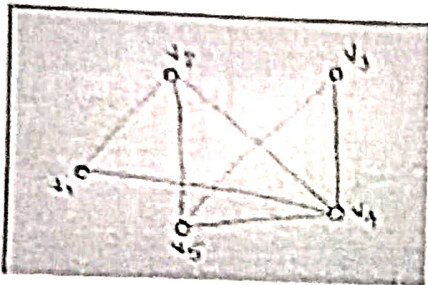
⑤

Write the circuit matrix of the following graph.



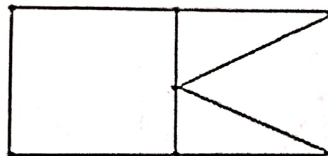
⑥

Find the adjacency matrix corresponding to the graph given by



⑦

a) Find at-least 6 circuits for the given graph and generate the corresponding circuit matrix representation with the circuits obtained. (Note: Assume suitable names for the vertices and edges.)



⑧

For the graph G below

a) List all ~~disj~~ circuits in G

b) Construct the circuit Matrix.

